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Review article

The dilemma of PID tuning[☆]

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ABSTRACT

A lot of automatic feedback control and learning tasks carried out on many dynamical systems still fundamentally rely on a form of proportional–integral–derivative (PID) control law. The PID law is often viewed as a simplistic computational control algorithm. However just like all non-convex optimization problems, tuning the PID algorithm for accurate and stable closed-loop control becomes a NP-Hard Problem. This leads to a dilemma, for both users and designers, most especially in practise. It is then no wonder that tuning software is a big business in the industrial automation sector. In this review, we present and classify PID tuning methods till date into three general areas. Finally, we then present a proposal to minimize the dilemma of complexity and cost that has become associated with tuning the three main parameters of the PID control law. Hopefully, continuous attempts at the minimization of this dilemma can lead to both a money-savings investment and a significant improvement in the field of PID control design.

1. Introduction

The controller in software is known as a control algorithm, and it is the brain of an automatic control system. Control system design theory is the special theory that studies how control algorithms are systematically designed and how to ensure their stable and accurate performance (Guo, 2020). One central concept in control theory is the feedback principle. The presence of feedback realizes a closed loop. The main aim of feedback is to deal with the influence of uncertain internal system and external forcing factors on the stability and accuracy performance of the measured output(s) of a system, commonly called the *process variable*. Consequently, a central motivation in controller design is the need for guaranteed stable and accurate closed-loop performance of dynamical systems, especially in motor-control tasks (See Figs. 1 and 2). Closed-loop performance in terms of both accuracy and stability is therefore critical. In this case, a well-designed feedback controller then becomes a real money-savings and life-savings investment (Yang, 2020). In other words, both optimality and more importantly robustness are important (Keel, & Bhattacharyya, 2016) in the face of uncertainty. Using machine learning lingo, what this means is that: compared to a controller overfitting on control tasks at a particular period of time, generalization of the controller to control tasks at all time intervals is more important. Feedback is therefore, important in automation and intelligent systems. In practise, a lot of these automation tasks still fundamentally rely on a form of Proportional–Integral–Derivative (PID) control (Abramovitch, Hoen, & Workman, 2008; Guo, 2020).

On this subject of PID control design, the recent book (Wang, 2020) is a recommended text for beginners. A discussion on PID tuning has been given in a recent review by Borase, Maghade, Sondkar, and Pawar (2020), where the authors classify PID tuning into classical and intelligent methods. In contrast, and more suitably, in this review, PID control design (tuning) methods are presented from the design perspective of being either plant-model based, plant-model free or an hybrid of both perspectives. Specifically, the focus of this paper is solely to give a concise review of the complexity and cost dilemma in tuning the PID control law, as observed in these approaches. This has led to a continuous, unrelenting search for novel and better methods than is currently available in theory for practical purposes. As emphasized in Díaz-Rodríguez, Oliveira, and Bhattacharyya (2015) and Koelsch (2014), despite the boom of industrial tuning software, most real-time PID controlled loops in operation still behave poorly after some time, and need retuning, often increasing the costs of regulatory-loop maintenance. It is, then, no wonder that recently Guo (2020) argues the need to review PID control design theory. Also, Aström, and Hägglund (2018) has suggested the need for more extensive work on automatic adaptive PID tuning algorithms.

In the next sections that follow, we will focus on the PID law, its tuning dilemma (costs and complexity) and then succinctly discuss the

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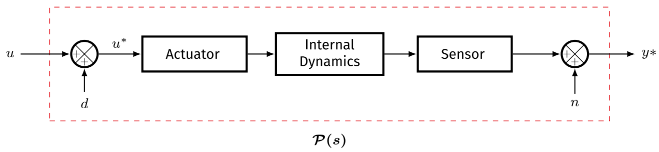

Fig. 1. Illustration of a Physical Dynamical System $\mathcal{P}(s)$, where d and n are the respective input and output disturbance and uncertainty lumped into the system.

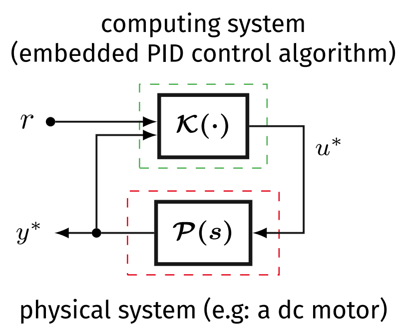

Fig. 2. A PID controlled loop or closed-loop system.

three categorized perspective of PID tuning methods with respect to the dependence of a plant model and the explicit use of numerical optimization techniques. A major pre-requisite for systematic control design is that some form of model is always needed for guaranteed control (Campestrini, Eckhard, Bazanella, & Gevers, 2017; Gevers, 1996; Van Den Hof, & Schrama, 1995). Subsequently, we conclude this review by proposing a theory to streamline the underscored costs and complexities of the time-consuming model identification and computational expensive optimization involved with PID tuning. In brief, we argue that the information of time, specifically the *closed-loop input–output settling-time* can be a sufficient system dynamics property for modelling feedback systems with a guaranteed finite settling-time. Therefore, for designing the parameters of the PID control-law, the development of automatic and robust methods for this type of settling-time identification becomes an open problem.

2. PID control law

Seminally, the work of Minorsky (1922) realized the PID law as a computational mimicry of the natural perception–action control principle deployed by skilled helmsmen. The PID law's structure for control is to use the past, present and estimated future error information (Bennett, 2001). Today, one general computational representation of the PID law is a two-degree-of-freedom (2DOF) control structure (Abdelaty, Ahmed, & Ouda, 2018; Araki, & Taguchi, 2003; Aström, & Hägglund, 1995; Viteckova, & Vitecek, 2015; Wang, 2012) aimed at simultaneous disturbance-rejection and set-point tracking. Such 2DOF structure with respect to the proportional and derivative set-point weights b and $c \in [0, 1] \in \mathbb{R}^+$ is

$$u = \mathcal{K}(r, \mathrm{sat}(y)) = K_p \left(u_p + u_i + u_d\right) \tag{1}$$

where

$$u_p = e_p, \quad u_i = T_i^{-1}\, e_i, \quad u_d = T_d\, e_d \tag{2}$$

$$e_p = b\, r - y^* \tag{3}$$

$$e_i = \int \left(r - y^*\right) dt = \int e\, dt \equiv s^{-1} e \tag{4}$$

$$e_d = c\, \dot{r} - \dot{y^*} \equiv s \left(c\, r - y^*\right) \tag{5}$$

where $u^* = \mathrm{sat}(u)$ and $y^* = \mathrm{sat}(y)$ are the saturated input and output of an illustrative single-input single-output system respectively, such that the three PID contributing terms (or kernels) $u_p, u_i, u_d \in \mathbb{R}$ are related to the error terms $e_p, e_i, e_d \in \mathbb{R}$ respectively. $K_p \in \mathbb{R}$ is the proportional gain, $T_i \in \mathbb{R}$ is the integral time-constant, $T_d \in \mathbb{R}$ is the derivative time-constant, $e \in \mathbb{R}$ is the full error signal.

Modern PID Formalism. Considering that a defining factor on the natural perception–action control principle for learning, decision and prediction in living organisms is criticism. In a recent dissertation (Somefun, 2021), the PID law (1) was redefined in a critic form (6).

$$u = \mathcal{K}(r, sat(y)) = K_p \left(\lambda_p\, u_p + \lambda_i\, u_i + \lambda_d\, u_d\right) \tag{6}$$

where $\lambda_p, \lambda_i, \lambda_d \in \mathbb{R}^+$ are the *critic weights* on each of the three PID contributing terms related to the error terms respectively, such that $\lambda_p \equiv 1$, and $0 < \lambda_i \leq 1$, $0 < \lambda_d \leq 1$ represents the degree of belief (between 0 and 1) on each of the three PID contributing terms. The introduced critic weights serve as a belief-basis for discriminating the output integral and derivative contributions. This critic formulation (6) differs from the classic definition which assumes that the PID's three contributing terms equally dominate the output decision of the PID function, that is $\lambda_p, \lambda_i, \lambda_d \equiv 1$.

Motivations and Successes. Specifically, it can be stated that any controller or learning structure involving error feedback is a form of the PID law. This PID control law (1) is, both in theory and practise, specially regarded as ubiquitous (Aström & Hägglund, 1995; Johnson & Moradi, 2005; Samad, 2017; Wang, 2020). A recurring motivation for this is because, although an apparently simple, yet powerful mechanistic representation of the universal error feedback law (Silva, Datta, & Bhattacharyya, 2005; Vagia, 2012), PID control applied to many practical problems continues to provide simple, least-cost, and at-least satisfactory robust control performance (Bucz, & Kozáková, 2018; Li, & Wang, 2016; Peretz, 2018; Wang, 2020; Yu, 2018).

More recently, the PID law has been applied as an optimizer in the weight-training of deep neural networks (Wang et al., 2020). It is often represented or designed as a continuous-time system but implemented as a discrete-time system (that is in software) (Wang, Han, Liu, & He, 2020). In retrospect, the PID law is simply an algorithm or a computer, that can be realized as software or as an hardware.

The PID controller's behaviour is such that it assumes the presence of measured or estimated output state variables (output and its derivatives) for the control of one output in a system. Robust control theories like Time-Delayed Control (TDC) (Roy, & Kar, 2020) and Uncertain Disturbance Estimation (UDE) (Zhong, Kuperman, & Stobart, 2011) have been shown to reduce to a specific form of PID control (Chang, & Jung, 2009; Nam, 2016). These robust control theories have further shown that the PID control structure is generally an implicit closed-loop model reference structure for the control of systems with unknown (uncertain) nonlinear dynamics. It has also been noted in Aström and Hägglund (1995) that PID control appears to follow an implicit model. The use of such closed-loop adaptive model reference architectures proposed to deal with parametric uncertainties for guaranteed transient properties has been treated in Gibson, Annaswamy, and Lavretsky (2013) and Gibson, Qu, Annaswamy, and Lavretsky (2015).

Although, historically, the PID controller was first mechanically implemented by means of gears (Bennett, 2001; Nicolai, 1922), it is not a souvenir of the past century of engineering (Aström & Hägglund, 2018). Many researchers and authorities have however expected control design advances like fuzzy logic, neural network, model predictive control (MPC), dynamic matrix theory, reinforcement learning, dynamic programming, to render the PID obsolete (Aström & Hägglund, 2018; Johnson & Moradi, 2005). Analysing the state of optimal and robust control, this point is stated clearly in Keel and Bhattacharyya (2016) that in the last six (6) decades, the control community's success has been limited in the direction of obtaining robustness through quadratic optimization. One core reason given and proved in Keel, and Bhattacharyya (1997) and Keel and Bhattacharyya (2016) is that such design methods lead to high order, fragile controllers.

For reliability, the lowest level of all complex system architectures is still a PID regulatory loop (Katebi, 2007). Industry assessments still view the PID as a important core in loop regulation and PID tuning as an important problem (Koelsch, 2014; Samad, 2017). One reason for this, may be that the PID law is a representation of the natural law of feedback. Despite the nonlinear and varying dynamic characteristics of many physical processes, a well tuned PID gives a good tradeoff between robustness and performance. The PID currently enjoys popular use in the chemical, food, oil and gas processing, automotive, motion, electronic industries and many other fields (Diaz-Rodriguez, Han, & Bhattacharyya, 2019). It is widely respected in the industry and a good percentage of researchers in the control-theory community have in more than the last three decades contributed to research on PID control theory and industrial engineering practise (Datta, Ho, & Bhattacharyya, 2013).

However, the success recorded by the PID law as a controller is overshadowed by certain real-time design and performance issues. One, a large percent of PID controllers in operation are in manual. Two, from a study carried out by ABB, most of the PID loops operating in automatic behave poorly after some time than in open loop operation (Koelsch, 2014). Although, there is also some knowledge gap between industrial practices and theoretical advances, these performance issues can either be traced to being a result of poorly chosen operating PID parameters which are products of the selected PID tuning method or traced to adversarial variations (uncertainty) in the dynamics of the system under control from the identified nominal models used for control design (Diaz-Rodriguez et al., 2019; Koelsch, 2014).

Meanwhile, for context, before we discuss these problems on the design of the PID law, we need to consider the mathematical language of *differential equations* used to model the PID law as an artificial representation of natural human perception–action negative feedback-based intelligence.

3. Representations: Integer order or fractional order

Differential equations (DEs) are used as the universal computational language for the formal description of dynamical systems (Li, Zheng, & Wang, 2019). The general form of the conventional integer-order DE is the fractional-order DE. On this subject, there is an ever-growing body of research works, particularly in the control literature on fractional-order PID (FOPID) controller representations as opposed to the integer-order representation of the complex Laplace variable given in (4) and (5). As proposed by Podlubny (1999), in the fractional from, these equations become

$$e_i = \mathbf{D}^{-\mu} e(t) \equiv s^{-\mu} e \tag{7}$$

$$e_d = \mathbf{D}^{\delta} \left(c\, r - y^* \right) \equiv s^{\delta} \left(c\, r - y^* \right) \tag{8}$$

where the fractional variables $\mu, \delta \in \mathbb{R}^+$. The growing interest in fractional representations, has been attributed to the extra degree of tuning freedom, allowing for frequency response shaping, given by the fractional variables (Dastjerdi, Vinagre, Chen, & HosseinNia, 2019; Tepljakov et al., 2021; Vinagre, Monje, Calderón, & Suárez, 2007). Therefore it is often claimed that using FOPIDs give rise to superior control loops. However, there are certain aspects left to be critically addressed (Tepljakov et al., 2021). One, it is still difficult to concretely validate the cost–benefit advantage of using fractional representations for real-time PID control, as they are more computationally expensive in terms of memory requirements and speed of algorithm execution on a computer chip (Petráš, 2012), hence subtly introducing delays in the control loop. This complexity of both tuning and implementation has made its adoption not widespread in the control engineering industry (Dastjerdi, Saikumar, & HosseinNia, 2018; Dastjerdi et al., 2019; Vinagre et al., 2007). It remains to see if in the future such bottlenecks can be removed. Two, in the literature, real-time implementation of fractional integrals and derivatives are always usually, approximated to equivalent higher integer-order linear filter or transfer-function representations (Dastjerdi et al., 2019). This then begs the question, of its claimed advantage over integer-order representations.

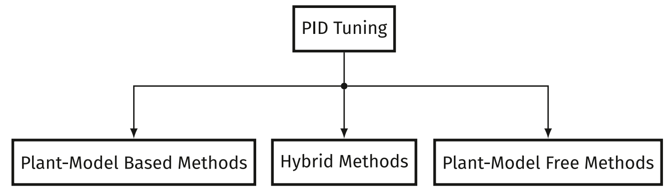

Fig. 3. General classification chart of PID tuning methods.

Furthermore, some questions remain as regards the comparative performance analysis of tuned fractional PID representations to integer-order representations, as shown in works like that of Koszewnik, Pawłuszewicz, and Ostaszewski (2021). Most tend to use a sub-optimal tuning method to find the PID parameters for the integer-order forms. Is it possible that the use of other tuning methods could have provided similar or better performance compared to the fractional forms, as reported in the literature, such as in Bingi, Ibrahim, Karsiti, Hassan, and Harindran (2018)? This question is important, as the search space for finding even the best three gain terms of integer PID forms is an NP-hard problem.

Again, it is possible to obtain better performance from PID controllers by considering the PID representation from a *signal processing perspective of filtering*. The integral and derivative terms can then represent integer-order filter systems of higher order integrals and derivatives (Note that the benefit of higher order derivatives greater than order 2 is not always clear in terms of control performance), a notion close to that proposed in (Huba, 2015; Huba, & Vrančič, 2018). However, as fractional representations are always realized as linear high dimensional integer-order filter transfer-functions, they have the advantage that the increasing amount of filter parameters is connected to basically just two fractional parameters which can be tuned more easily. To conclude, currently, control engineers are still faced with a dilemma in choosing fractional representations over integer-order representations of the PID. For real-time controller realization in software for fast algorithmic execution and safety-critical control purposes, the use of integer-order representations remains the dominant choice, as it reduces the PID's computational and tuning complexity.

It was Wiener (1950) who theorized that: nature and specifically, humans use a negative-feedback, retuning (adjustment) process to gain new knowledge and achieve goals. This way, feedback control theory can also be viewed as *learning theory* or the *science of intelligence systems* (Bhaya, & Kaszkurewicz, 2007; Hershberger, 1990; Poljak, & Tsypkin, 1973; Tsypkin, 1971). Arguably, the most successful application of Wiener's cybernetic theory of artificial intelligence goes to the PID algorithm, used in different forms as gradient-descent optimization algorithms (Hu, & Lessard, 2017; Lessard, Recht, & Packard, 2016; Recht, 2019; Wang et al., 2020; Zweiri, Seneviratne, & Althoefer, 2005) for learning (optimizing or controlling) the weights of artificial neural networks, based on the reconstruction errors of the neural network (LeCun, Bengio, & Hinton, 2015; Schmidhuber, 2015; Werbos, 1975). However, as elegant as the representations of the PID are for mimicking intelligence, unfortunately, certain difficulties arise in its design for control, also known as tuning. These problems has made it that no one method can be said to be universal or best. To date, we are still lacking knowledge on the best way to achieve a universal optimal solution to the tuning design of the PID law, whether as a controller or an optimizer.

4. PID tuning problem

Interestingly, one would have thought that the seminal work in Ziegler, and Nichols (1942) on *Optimum settings for automatic controllers* would be the final and universal answer to PID tuning. On the contrary, significantly, the design problem (tuning) of PID controllers has attracted unceasing theoretical and practical research interest (Aström & Hägglund, 2018; Vilanova, & Visioli, 2012; Wang, Yan, Li, & Sun, 2018). To achieve best performance in terms of stability and accuracy, the single best control design solution to tuning a closed-loop system controlled by a PID (which can be termed a *closed PID-loop*) remains an open question in many applications (Guo, 2020; Saab, 2017; Smithm, 2018a). One reason for this is the absence of perfect models for describing real-world systems which are inherently uncertain systems (Ang, Chong, & Li, 2005; Dastjerdi et al., 2018; Jantzen, & Jakobsen, 2016; Sung, Lee, & In-Beu. Lee, 2009).

The systematic selection of the optimum set of PID parameters have therefore been categorized as an NP-hard problem in terms of complexity (Koszaka, Rudek, & Pozniak-Koszalka, 2006). This implies there is no unique solution. It also means that tuning for guaranteed accurate and stable control performance can be burdensome even for very common servomechanism applications (Grimholt, & Skogestad, 2018; Killingsworth, & Krstic, 2006; Roux-Oliveira, Costa, Pino, & Paz, 2019).

Indeed, the literature as evident in Vilanova and Visioli (2012) on PID design (systematic tuning) for single-input single output (SISO) and multiple-input multiple-output (MIMO) systems is rich and extensively wide. These methods use one or more of classical control theories or optimal, robust and adaptive control theories, which are subsets under nonlinear control theory (Silva et al., 2005; Slotine, & Li, 1991). Earlier surveys of fixed and adaptive PID tuning methods have been presented in Aström, Hägglund, Hang, and Ho (1993), Cominos, and Munro (2002) and Koivo, and Tanttu (1991). Similarly, in Aström and Hägglund (2018) a more recent overview on the place of PID control design as an active research in adaptive control (Black, Haghi, & Ariyur, 2014) and the need for automatic tuning algorithms instead of simple rules was presented.

A major pre-requisite for systematic control design is that some form of model is always needed for guaranteed control (Campestrini et al., 2017; Gevers, 1996; Van Den Hof & Schrama, 1995). This is known as system identification (Aström, & Wittenmark, 1971; Gevers, 2005; Ljung, & Vicino, 2005), where a computational model is explicitly identified either by first principles of physical laws or either explicitly or implicitly from the experimental input–output data of a dynamic system, also known as *training data* in the machine-learning literature (Ljung, Chen, & Mu, 2020; Pillonetto, 2018). This aim of identifying models from data apart for control purposes may be for simulations. The practice of system identification has therefore shaped the way controllers are designed in order to achieve certain closed-loop performance metrics.

In this context, the methods of tuning PID controllers can therefore be categorized (See Fig. 3.) into three general perspectives based on the concept of a model for describing or representing either the closed-loop system or the open-loop system to be controlled. Namely: Plant-model based methods, which require identification of a plant-model structure and knowledge of its parameters for tuning. These methods may or may not be data-driven; Plant-model free methods, which are purely data-driven approaches with optimization for tuning; and Hybrid methods, which combine a kind of priori plant-model knowledge with data-driven approaches for tuning.

Also, it should be noted that tuning can be done offline or online. When tuning is done online, we mean that the controller is tuned while it is providing the control function in a closed-loop. When tuning is done offline, we mean that the controller is tuned while it has not yet started providing the control function in a closed-loop. Further, it should be noted that controllers are tuned on the fundamental basis that power supply to the physical dynamic system (including the chip hosting the controller, its actuators and sensors) is averagely always constant and reliable. In practise, the degradation of power supply such as in battery-powered systems, means set-points cannot be reached, disturbances cannot be rejected and hence failure in control.

In detail, we continue the discussion of these categorical methods and their shortcomings in the following sections.

5. Plant-model based methods

Notably, a greater percentage of PID control design methods (fixed and adaptive) available are highly dependent on the knowledge of derived mathematical model approximations from first principles or fitted mathematical model approximations of the actual underlying physical dynamical system from experimental data (Astrom, & Hagglund, 2006; Aström, & Hägglund, 2017; Ellis, 2012; O'Dwyer, 2009).

In modern terms, considering trade-offs between performance and robustness (Garpinger, Hägglund, & Åström, 2014; Grimholt & Skogestad, 2018; Segovia, Hägglund, & Åström, 2014), the identified parameters of these plant models and structures are explicitly used as design constants to set the PID parameters (Levine, 2011; Li & Wang, 2016; Skogestad, 2006; Sung et al., 2009; Visioli, 2006). Corresponding adaptive methods achieve parameter estimations using derived plant model parameters. For LTI (Linear Time-Invariant) MIMO systems, such tuning methods proposed to guarantee robust closed-loop stability can be found in Cao, and Chen (2014), Gundes, and Ozguler (2007) and Saab, and Toukhtarian (2015). In this category, some methods like in Reynoso-Meza, Garcia-Nieto, Sanchis, and Blasco (2013) employ multi-objective optimization for tuning. These methods generally involve minimization of some performance index. The use of Lyapunov stability and linear matrix inequalities (LMI) analysis also form the basis of some other tuning methods (Mendoza, Zavala-Río, Santibáñez, & Reyes, 2015).

Other modern methods in the category mix of plant-model knowledge and optimization objectives include: the general model predictive control (MPC) (Klaučo, & Kvasnica, 2019), H-infinity (Diaz-Rodriguez et al., 2019), quantitative feedback theory (QFT) (Comasòlivas, Escobet, & Quevedo, 2012; Mercader, Åström, Baños, & Hägglund, 2017) and linear quadratic Gaussian (LQG) control methods.

As most engineered dynamical systems are cyber–physical systems with actually complex, unknown dynamics and large time-varying time constants (Wang, Davison, & Davison, 2013). From a linear dynamical systems perspective, a plant-model based approach by system (or process) identification, such as the first-order plus time delay model is useful and more so, reasonable in some sense (Bai, & Roth, 2019; Forrai, 2013; Lynch, Marchuk, & Elwin, 2016; Samad, 2017; Samad, & Annaswamy, 2013). Notwithstanding, these models are imperfect representations of the physical systems, which although may operate in a linear sense, are actually both non-linear and uncertain (Guo, 2020). Therefore, these plant-model based approaches may not fully exploit the best quality of control accuracy using the PID. According to a ASEA Brown Boveri (ABB) survey in Koelsch (2014), after some time these tuning methods, which are commercially available become unreliable. Also, it has consistently been reported in the case of process industries, that such identification, although useful for testing, can be a time consuming and expensive process (Jantzen & Jakobsen, 2016; Roux-Oliveira et al., 2019).

A summary list of the general shortcomings associated with plant-model based methods include:

1. imperfect models compared to the actual uncertain, complex plant dynamics
2. significant retuning may be frequently needed
3. time-consuming identification process
4. usually fixed tuning rules, but most modern methods make heavy use of numerical optimization techniques

6. Plant-model free methods

Therefore, as the challenges of plant-modelling being time-consuming, costly and complex or good plant models being mathematically unavailable surrounds process identification for control. This has motivated interest in design methods that are not plant-model based. Deviating from model-based conventions are model-free methods, which can essentially be categorized as purely data-driven control design methods.

In this case, what is often meant, is the absence of a mathematical model of the plant for tuning the controller parameters and that under certain assumptions, there is no explicit identification of the parameters of a certain model structure from data.

These methods for tuning PID gains are usually based on a reference model from closed-loop input–output data experiments and minimization of some objective function (or performance index). A notable feature of plant model-free methods, is that they realistically assume that the true system is inherently nonlinear and uncertain.

Among them, one popular technique is the Relay Feedback Test (RFT) (Boiko, 2013), specifically the Relay Auto-tuning method (Aström & Hägglund, 2018). Relay Auto-tuning, uses modified Ziegler–Nichols (Z–N) tuning rules to automate the frequency response identification of a dynamical system (gain and phase margin information). Although, most recent variants involve multi-objective optimization, the RFT method still involves some form of process identification and sometimes the use of plant-model knowledge. Primarily, this approach relies on describing nonlinear function analysis, and so frequency limit cycles are common. The accuracy of the identified critical frequency point of oscillation is also still a subject of active research (Hornsey, 2012; Zeng et al., 2019).

Particularly notable in the model-free category, are gradient-based design methods which explicitly minimize a cost function (performance-index) for real-time optimization, such as the unfalsified method, iterative feedback tuning (IFT) (Ho, Hong, Hansson, Hjalmarsson, & Deng, 2003), iterative learning control (ILC) (Moore, & Xu, 2000), and extremum-seeking (ES) method summarized in Killingsworth and Krstic (2006). In Paz, Oliveira, Pino, and Fontana (2020) and Roux-Oliveira et al. (2019), through ES, the gains of the PID control law were adaptively tuned for functional neuromuscular electrical stimulation, specifically to control and coordinate the positioning of the arm–elbow joints of stroke patients. The assumption in this case was that the neuromusculoskeletal system is input-to-state stable (ISS). This implies that for any bounded control input, both the state and output variables are also bounded.

On the other hand, for the IFT, usually multiple offline closed loop experiments have to be carried out (Lequin, 1997) to estimate the objective function's gradient. The ILC approach leverages repetition, but to generally achieve perfect tracking performance, it requires physical systems to satisfy an identical initialization condition, which is a very restrictive condition (Guan, Zhu, Wang, & Liu, 2014; Preitl et al., 2007). Another is Virtual Reference Feedback Tuning (VRFT) (Cao & Chen, 2014; Formentin, Savaresi, & Re, 2012) or fictitious reference iterative tuning (FRIT) (Ashida, Wakitani, & Yamamoto, 2017; Kano et al., 2010; Tasaka, Kano, Ogawa, Masuda, & Yamamoto, 2009; Wakitani, Yamamoto, & Gopaluni, 2019). An alternative model-free learning (optimization) framework to ES and IFT for LTI MIMO systems (assumed unknown) was proposed in Wang et al. (2013), which also involves iteratively carrying out closed-loop experiments. There is still however no guarantee that the tuned gains will always approach the optimum.

Likewise, a summary list of the general problems associated with plant-model free based tuning methods include:

1. possess restrictive design conditions for systems in practise.
2. more computationally expensive, and sometimes unreliable error-minimization optimizations on data.
3. time-consuming data-driven experiments to ensure data quality
4. use of, sometimes, complicated objective-function formulations
5. since no explicit model is used, some trial and error values, or simulations may be needed to arrive at satisfactory settings.
6. nominal control is only as good as data, data may not capture all actual operating conditions, so actual control may be sub-optimal.
7. may require a try-test-adjust approach in setting the values of certain hyperparameters used for tuning.

7. Hybrid methods

Hybrid methods cover methods that combine both the use of a form of plant-model knowledge (not necessarily a parametric plant-model structure) with data-driven plant model-free approaches for tuning.

In this category, there are methods, that depending on the way control techniques (optimal control methods, adaptive control methods, pole-placement methods, computational-intelligence methods) are used, could be classified as using both or one of plant-model based and plant-model free characteristics for tuning.

Furthermore, there are other cases, where, despite being significantly plant-model free, some form of implied model or characteristic of the system from observed data is used to directly tune the PID controller parameters, whether adaptively or in a fixed form.

In the first case, a lot of recent works on PID tuning design employ optimal control techniques, and tend towards mixing different robust design objectives, such as specification of gain and phase margins, by employing constrained optimization and multi-objective optimization. Particularly, considering the attenuation of measurement noise (filtering) and time-delay, different tuning approaches and studies such as in Larsson, and Hägglund (2011), Mercader et al. (2017), Sekara, and Matausek (2009) and Veronesi, and Visioli (2018) have been carried out.

Again, a notable method in this category, is the dominant eigenvalue assignment, crudely known as pole-placement methods. Such approaches, of which a greater percent are plant-model based, are treated in works such as Datta et al. (2013), Diaz-Rodriguez et al. (2019), Han, and Bhattacharyya (2018), Keel and Bhattacharyya (2016), Srivastava, and Pandit (2016), Wang, Han, Liu, and He (2020) and Zítek, Fišer, and Vyhlídal (2013). An attracting point of this method, is that performance specifications such as stability and dynamic performance directly correlate to the assignment and distribution of dominant closed-loop eigenvalues (poles). The added use of multi-objective optimization is also very common in some of these approaches.

In addition, more recently, there is a trend, observed in works such as Almabrok, Psarakis, and Dounis (2018), Ekinci, and Hekımoğlu (2019), Ekinci, Hekımoğlu, and Izci (2021), Hekimoğlu (2019), Izci, and Ekinci (2021) and Mandava, and Vundavilli (2019) which propose metaheuristic (evolutionary) optimization algorithms as new PID tuning methods. These methods consider multi-objective optimization, using evolutionary optimization and specification of performance constraints such as stability, bandwidth, and robustness. Although, these methods are often posed as model-free methods, in their design for control purposes, they still often require the need of a plant model of the system to be controlled for simulation purposes. At the basic level, they can be viewed as stochastic search methods. Although promising, they have not found wide use in the process industry, since any serious uncertainty in the parameters of the system can lead to instability, and, in some cases, the converged solutions are less than optimal (Dastjerdi et al., 2018; Li et al., 2019). Also, sometimes, before they are executed, they depend on some initial settings from another tuning method as good starting-points for optimization. Some recent attempts at practical implementations of such techniques can be found in Ekinci, Izci, and Hekımoğlu (2021) and Chen, Bowels, Zhang, and Fuhlbrigge (2019).

In the second case, one notable method, was proposed in a study by Jantzen and Jakobsen (2016). Originating from fuzzy control study, the central argument in this approach is that the time-consuming system identification of plant-models for PID tuning can be reduced to considering the physical meaningful performance specification of desired settling-time based on operator knowledge and system operating behaviour. Also, more recently, Zhong, Huang, and Guo (2020) proposed a tuning method based on a parametric bandwidth formula of the extended state observer (ESO), which uses pole-placement and model-reference following paradigms to connect the active disturbance rejection control (ADRC) concept to design the PID law.

The use of explicit reference models, is common in tuning methods which use ideas of model following adaptive control. Examples are direct (or indirect) model reference adaptive methods and existing self-tuning methods (Aggarwal, & O'Reilly, 2006; Bobál, Böhm, Fessl, & Machácek, 2005). Most of these methods usually suffer from poor transient performance or (and) may involve some expensive online parameter optimizations and estimations.

Furthermore as hybrid methods, tuning approaches that exploit the artificial intelligence theories of fuzzy-logic, artificial neural-networks, and reinforcement learning for PID design exist (Jafari, & Dhaouadi, 2011; Kofinas, & Dounis, 2019; Malekabadi, Haghparast, & Nasiri, 2018; Marino, & Neri, 2019; Mendel, 2017; Pirasteh-Moghadam et al., 2020; Savran, & Kahraman, 2014; Shipman, & Coetzee, 2019; Srivastava et al., 2018; Taeib, & Chaari, 2015; Wang, Cheng, & Sun, 2007). These methods have generally been applied to the control of systems with the assumption that they are inherently nonlinear. Fuzzy-logic PID control techniques fuzzify the system error states, however, it is afflicted by time-consuming rule parameter optimization problems (Bai & Roth, 2019). On the other hand, deep learning techniques involving neural networks and reinforcement learning methods have been noted to often give poor results in real-time control practise (Retch, 2020).

Again, a summary list of the general problems or characteristic issues associated with hybrid tuning methods include:

1. dominant use of different types of heuristics or optimization techniques
2. inherits some problems from its two basis methods, so tuning solution may be less than optimal and cause instability
3. poor transient performance, if real-time
4. expensive real-time parameter optimizations, may give poor results in practise.

Therefore, having considered these categories, we see a trend in the use of numerical optimization techniques in each of these general categories. In the next section, we move on to discuss the effect of this from the perspective of control software use-case in real-time, as opposed to simulation time.

8. Tuning software and numerical optimization

In Zhang, and Guo (2019) and Zhao, and Guo (2020), the authors propose new universal theorems for the design of the PID law focused on dynamical systems as nonlinear and uncertain. The authors highlight that precise understanding and rationale for the remarkable practical effectiveness and capability of PID control is still limited. This view is affirmed by statistics from industrial process automation, which show that: although the continued evolution of hardware and software for discrete-time control has advanced the use of PIDs, the weakness of poor tuning is still prevalent (Allan, 2018). Interestingly, a reactive effect of this is the requirement of expensive tuning software. This may explain why commercial tuning software is big business in the control industry today (Koelsch, 2014).

Most auto-tuning software are heavily based on either one-shot or real-time plant-model fitting and the use of numerical optimization techniques. A downside to this approach is that since actual physical systems deviate from this estimated models. They tend to embody some form of nonlinearity, noise and unpredictable perturbations. Therefore, inaccurate estimations through such tuning software may lead to closed-loop performance degradation (Allan, 2018). The effects of such poorly tuned closed PID-loops have been noted to include: increase in process variability, increased energy consumption, operating costs and reduction or unimproved product quality.

Therefore, despite the reliance on numerical optimization present in tuning methods and their implementation as software, many physical dynamical systems still fall back to being tuned manually or heuristically in practise (Smith, 2018b). The complexity of this task is evident in the search-space of numerical optimization functions and the number of PID gains to tune as the number of controlled output variables increase. For instance, in the control of a system with 6 motors, this would mean simultaneous tuning of at least 18 parameters or 24 parameters in the case of added filter constants. It is believed that this complexity can be reduced to some minimum level. As such, adaptive but robust PID control design will continue to attract research interest.

Specifically, alternative tuning frameworks for streamlined control in the closed-loop data-driven category, without the overhead of time-consuming plant-model identification and multi-objective numerical optimization, would help to improve the PID's widespread adoption and contribute to better product quality (Chen et al., 2019; Jantzen & Jakobsen, 2016), hence reduction in the costs and complexities encountered with PID tuning.

9. Future: Dynamical system description for Control by the Information of Settling-time

Accordingly, there needs to be a renewed interest in the search for automatic adaptive tuning algorithms that can help streamline the costs of PID tuning and reduce the complexities involved. Several relatively recent works such as in Bucz and Kozáková (2018), Jantzen and Jakobsen (2016), Nguyen, and Nguyen (2018), Viteckova and Vitecek (2015) and Wang et al. (2018) have all emphasized the important consideration of settling-time for both streamlining and improving PID tuning.

In an attempt to resolve this PID tuning dilemma, a novel proposal may be to consider the **Closed-PID loop model (CPLM) Following Control (CPLMFC) Theory**.

CPLM. First, the PID law by its structure is a controller with its own internal and implicit closed-loop reference model that can be called the closed-PID loop model (CPLM), which is the implicit ideal closed-loop response that it tries to reference.

Low-frequency Bandwidth. Second, the CPLM can be viewed as a specific case of dominant eigenvalue assignment of complex-conjugate poles through the specification of its low-frequency bandwidth.

Closed-loop Settling-time. Third, the closed-loop settling-time in connection to the low-frequency bandwidth can be used a sufficient systems dynamics description for control. This way, settling-time identification then becomes a robust test for the maximum capability (limits) of a feedback system (Guo, 2020) as observed in the time-domain. Interestingly, the frequency (bandwidth) parameter and the finite settling behaviour (FSB) (Rao, & Bernstein, 2001) of a dynamical system are connected. Using the low-frequency bandwidth parameter, implicit dominant eigenvalues of the CPLM can be assigned through the integral and derivative(s) time-constants to ensure the actual closed PID-loop system stably and accurately follows the CPLM. The proportional gain could then be constrained to be an adaptive parameteric estimation of the control input(including disturbances)-to-output gain, ensuring global stability and accuracy of the overall feedback system.

Therefore, the implication of the CPLMFC Theory is that the PID law can in real-time adaptively tune itself using the CPLM in connection to an identified settling-time of the closed-loop system. An advantage of such a tuning framework, is that it requires no knowledge of the parameters of a plant (dynamical system) model structure or its order, and no explicit minimization of an objective-function for optimization. This theory was recently introduced in a master's dissertation (Somefun, 2021) and applied to the real-time adaptive (online tuning) control of a differential-drive mobile robot. Note that, a preliminary form of the CPLMFC theory was applied in Somefun, Akingbade, and Dahunsi (2020) for the robust speed-control of dc-motors.

However, this brings up the issue of how to reliably identify such a closed-loop settling-time in an automatic manner and online manner, directly from data, that is by real-time signal processing, instead of manually or offline. Again, with the adaptive nature (online tuning), of such a framework, comes the need for safety guarantees.

Therefore, in the CPLMFC framework, the need for a robust online settling-time identification process becomes very important. The development of such robust online algorithms and methods for automatic identification of the closed-loop input–output settling-time as a sufficient system dynamics property, then becomes a necessary problem to be solved.

Limits of Feedback and the Information of Time. Again, the information of time, has important effects on a closed-loop system. Since feedback can generally be viewed as a causal function of the observed output from a dynamical system (Guo, 2002). Its maximum performance is constrained to the limitations of the closed-loop's actuator, sensor and the time taken to execute the control-input. Therefore, the test for the maximum capability of a dynamical system, is to simply probe what it cannot do. In the time-domain, the limits of such performance is the system's settling-time. In other words, a system (closed or open) cannot react or stabilize itself faster than its dynamics allow. An attempt to violate this, especially in practise, often leads to excessive oscillations and overshoots in the time-response of such systems.

Settling-Time. To further understand the above statements, we recall that practical, canonical physical dynamical systems are inherently energy dissipating or storage systems (Tewari, 2002). On the application of an arbitrary step input, the output of such systems do not instantaneously take a steady-state value, but takes some amount of finite time to settle to an average value. This input–output delay-time interval to settle to an average value is a characteristic property of all controllable and stable dynamic systems. This finite input–output delay time interval, can be regarded as the settling-time, being the aggregated time delay dynamics that captures the input–output transition of such systems.

Ultimately, the settling-time embodies the observable performance indices of a control system's reliability, which is in the time-domain (that is: how rapidly a fluctuating measured process output takes to settle, and with what amount of maximum overshoot the measured output takes to settle to become exponentially close (accurate) in value to the desired process output or goal) (Atherton, 2015).

Dead-Time. Also, the interval of time known as settling-time is a superset of the dead-time (transport-lag) known by control researchers and engineers to cause difficulty in PID tuning, and so, degraded, non-smooth, limited regulation of feedback systems.

The first known timing effect, as seen during open-loop operation, is the inherent physical constraint that does not allow the observed output to change from its previous value until some finite period of time after the change in the control input. Nothing can be done about this.

The second known timing effect, as seen during closed-loop operation, is that it degrades the closed-loop transient, simply because with respect to time, the closed-loop is trying to control the observed output faster than its realistic limits. Targeting dead-time control, there exists significant quantity of research since the early periods of the twentieth century (Normey-Rico, 2007).

Finally, the third timing effect, is in the case of the sampling-time (Clair, 1993; Kerrigan, 2015; Lequin, 1997) to be discussed next.

On Sampling-Time. To conclude this section, it seems useful to have a short note on sampling-time τ_s for control.

Basically, there are at least two sampling-times used in realistic control design such as illustrated by (9). Hence, as opposed to most simulations, a real-time feedback system is viewed as a multi-rate discrete-time system. First is the open-loop sampling-time, the rate at which actuator (process input) and the sensor (process output) variables are sampled and measured. Next is the feedback (or control-loop or closed-loop) sampling-time that specifies how often the controller samples the measured process variable and then computes and transmits a new control input signal to the actuator. There is also the computational dead-time, which is recommended to be a very small negligible time.

$$\tau_s^{(y)} = l\,\tau_s^{(u)}, \quad \text{where: } l \geq 100 \quad \text{(recommended)} \tag{9}$$

Consequently, the minimum dead-time, in a control loop is the loop's sampling-time, and the maximum dead-time is the sum of all delay-time intervals in the loop, from the output to the input. Since, controllers sample observed measurements, act, then wait until next sampling-time, and also, communication protocols and scheduling operations of the embedded operating system contribute additional layers of dead-time and uncertainty to the overall system. Therefore, the controlled system or process has sampling-times of the controller's processor and communication protocols built naturally into its structure as dead-time intervals.

Fast and slow are relative terms defined by the inherent dynamics of the system (including its actuator, sensor, and processor). This is why selecting a sampling period (not too fast and not too slow) that satisfies the scheduling and control requirements is such an important and difficult task.

Sampling slower than the system allows, will have a negative impact on performance, like loss of data fidelity, introduction of extra dead-time. Sampling faster than the system allows, will have a negative impact on performance, like unnecessary computational overload and overhead, higher demand on high-end sensing, actuating and computing resources, with no performance improvement.

The control engineer would need to choose a sampling time such that data can be sampled, processed, and transferred by the software within the limits of the real time system's or hardware's working frequency. Anything too more or less will degrade the stability margins (and response) of the system.

Therefore, the achievable accuracy and stability of a PID controlled physical dynamical system is strongly related to time. Actuating faster or slower than the maximum time (or bandwidth) dynamics generating a signal leads to false reconstruction or observations and hence poor control. This is also a dilemma (Aström, & Wittenmark, 2013; Bai & Roth, 2019; Franklin, Powell, & Emami-Naeini, 2019) especially for tuning the PID controller in real-time. One recommended best practice, is that the control loop's sampling-time should be at least 10 times faster than the sampling-time of the measured process. There is still no universal rule, only heuristics like (9).

The relation of the information of time to tuning a feedback-controlled system is therefore, arguably, the most important.

10. Concluding remarks

This paper has provided a succinct review of the rich and persistent literature on tuning PID controllers, also known as PID control design. The review started with an introductory discussion on feedback theory in relation to the PID control law and its representations. Then the PID tuning problem was discussed next. This led to a general classification of available PID design methods into the following three perspectives: plant-model based, plant-model free, and hybrid methods. The review also highlighted certain generic problems with the methods discussed, as encountered in practice when tuning the PID law for controlling real-world physical dynamical systems. Next, the connection between tuning software and numerical optimization methods was discussed. Finally, closed-loop settling-time identification for control was proposed as a sufficient dynamical-systems descriptor to replace traditional plant-model parametric identification for control design.

Also, we remark that compared to using numerical optimization techniques on the PID law, there is a need for the control community to start viewing the PID as a natural optimization algorithm. In any case, it remains to see if, in the foreseeable future, there will be a universally acceptable adaptive and robust solution to the dilemma faced with the automatic tuning of a PID controller for long-term operation.

Because of the popularity of the PID in practise, future successful attempts at resolving its tuning dilemma can lead to significant money-savings investment and technological revolution in the engineering field of control design, learning theory and optimization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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